Math-Informed Reinforcement Learning for Optimal UAV Relay Positioning in 6G Networks

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***Abstract*—Deploying unmanned aerial vehicles as relay stations can extend coverage and increase throughput in wireless networks. Finding the best three-dimensional position for these aerial relays is difficult because the underlying optimization is non-convex. We present a math-informed reinforcement learning method called Successive Convex Approximation-Guided Actor-Critic (SGAC) that combines classical optimization with deep learning. SGAC uses analytical gradients from convex relaxations to initialize policy learning and guarantees performance at least as good as the classical baseline. Our experiments show throughput gains of 264% over random placement and 81% over analytical heuristics across 50 urban scenarios. The framework reaches 95% of its final performance within 50 training episodes, roughly four times faster than standard RL methods.**

***Index Terms*—UAV communications, 6G networks, reinforcement learning, convex optimization, relay positioning**

I. Introduction

Data-hungry applications and the growth of connected devices push wireless infrastructure toward new deployment models. Fixed base stations serve their purpose well but cannot adapt quickly to changing patterns of user demand. Unmanned aerial vehicles carrying communication equipment offer a flexible alternative, repositioning on demand to fill coverage gaps or handle traffic spikes.

Positioning these aircraft is harder than it looks. Unlike ground towers, a UAV can occupy any point within its operating envelope. The relationship between position and network throughput involves logarithms and minimum operations that create a non-convex landscape with many local peaks. Standard gradient methods often get stuck.

Successive convex approximation and similar techniques can navigate this landscape methodically, but they require several iterations per decision and may still settle for a local maximum. Reinforcement learning offers another path—agents that learn positioning policies from experience—yet typical RL algorithms need many samples before converging and provide no safety net if the learned policy performs poorly.

This paper bridges these two approaches. We observe that classical optimization yields not just a position estimate but also derivative information that points toward better solutions. Our SGAC algorithm treats the SCA output as a starting point and trains a neural network to predict small corrections. If the correction helps, we apply it; otherwise, we fall back to the SCA answer. This design guarantees performance at least matching the classical method while allowing the agent to discover improvements that SCA misses.

We make three contributions:

- A residual architecture where the policy learns adjustments to SCA solutions rather than raw positions, ensuring the learned policy never underperforms SCA.
- Analytical throughput gradients used as state features to speed learning.
- Theoretical results on reward alignment, Lyapunov stability, and sample efficiency.

II. System Model

A. Network Layout

Consider a downlink with one ground base station, K mobile users spread across a service region, and one UAV relay. The base station sits at fixed coordinates $\mathbf{q}_{\mathrm{BS}} \in \mathbb{R}^3$. User locations $\{\mathbf{q}_k\}_{k=1}^{K}$ follow an urban mobility distribution. The UAV occupies position $\mathbf{p} = [x, y, z]^\top$ subject to altitude bounds $z_{\min} \leq z \leq z_{\max}$ set by regulations and hardware limits.

B. Channel Characteristics

Air-to-ground links see both line-of-sight and obstructed paths. The probability of each depends on elevation angle and environment type. Path loss between the UAV at $\mathbf{p}$ and user k follows

$$L_k(\mathbf{p}) = 20 \log_{10}\left(\frac{4\pi f_c d_k}{c}\right) + \eta_{\mathrm{LoS}} P_{\mathrm{LoS}}(\theta_k) + \eta_{\mathrm{NLoS}} P_{\mathrm{NLoS}}(\theta_k) \tag{1}$$

where $d_k = \|\mathbf{p} - \mathbf{q}_k\|$ is Euclidean distance, θ_k is elevation angle, and $\eta_{\mathrm{LoS}}, \eta_{\mathrm{NLoS}}$ capture excess loss.

User k achieves rate

$$R_k(\mathbf{p}) = B \log_2\big(1 + \min\{\mathrm{SNR}_{\text{BS-UAV}}, \mathrm{SNR}_{\text{UAV-}k}(\mathbf{p})\}\big) \tag{2}$$

with B the channel bandwidth.

C. Objective

We seek the position maximizing total throughput:

$$\begin{aligned} \max_{\mathbf{p}} \quad & \sum_{k=1}^{K} R_k(\mathbf{p}) \\ \text{s.t.} \quad & x_{\min} \leq x \leq x_{\max},\ y_{\min} \leq y \leq y_{\max},\ z_{\min} \leq z \leq z_{\max} \end{aligned} \tag{3}$$

Non-convexity from the log and min makes this problem challenging.

III. SGAC Framework

A. Residual Policy

Let $\mathbf{p}_{\text{SCA}}$ be the position from T SCA iterations. Our policy network produces a correction $\boldsymbol{\delta}$ so that

$$\mathbf{p} = \mathbf{p}_{\text{SCA}} + \boldsymbol{\delta}_\theta(\mathbf{s}). \tag{4}$$

Setting $\boldsymbol{\delta} = 0$ recovers SCA performance exactly. The network only needs to learn when deviating helps, which simplifies the task considerably.

B. State Encoding

We combine geometric features $\mathbf{f}_{\text{geom}}$ (relative positions, distances), channel features $\mathbf{f}_{\text{channel}}$ (SNR values), and the throughput gradient $\nabla_{\mathbf{p}} R_{\text{total}}$ computed analytically:

$$\mathbf{s} = \left[\mathbf{f}_{\text{geom}}, \mathbf{f}_{\text{channel}}, \nabla_{\mathbf{p}} R_{\text{total}}\right]. \tag{5}$$

Including the gradient tells the network which directions improve throughput, accelerating learning.

C. Reward

The reward balances improvement over SCA with a penalty for large corrections:

$$r_t = \frac{R_{\text{total}}(\mathbf{p}_t) - R_{\text{total}}(\mathbf{p}_{\text{SCA}})}{R_{\text{scale}}} - \lambda \|\boldsymbol{\delta}_t\|_2. \tag{6}$$

D. Performance Floor

Before returning a position we check whether it beats SCA. If not, we revert:

$$\mathbf{p}_{\text{final}} = \begin{cases} \mathbf{p}_{\text{SCA}} + \boldsymbol{\delta} & R(\mathbf{p}_{\text{SCA}} + \boldsymbol{\delta}) \geq R(\mathbf{p}_{\text{SCA}}) \\ \mathbf{p}_{\text{SCA}} & \text{otherwise} \end{cases} \tag{7}$$

This mechanism guarantees performance regardless of policy quality.

IV. Theoretical Results

We establish convergence guarantees for SGAC through three main results. Define the value function $V(\mathbf{p}) = \mathbb{E}\left[\sum_{t=0}^{\infty} \gamma^t r_t \mid \mathbf{p}_0 = \mathbf{p}\right]$ and let $\mathbf{p}^*$ denote the global optimum of (3).

Theorem 1 (Reward Alignment)**.** *Under the reward structure in (6), the expected cumulative reward satisfies*

$$\mathbb{E}_\pi\left[\sum_{t=0}^{T} r_t\right] = \frac{1}{R_{\textit{scale}}}\mathbb{E}_\pi[R_{\textit{total}}(\mathbf{p}_T) - R_{\textit{total}}(\mathbf{p}_0)] - \lambda \mathbb{E}_\pi\left[\sum_{t=0}^{T} \|\boldsymbol{\delta}_t\|\right]. \tag{8}$$

Proof. Expanding the reward definition and summing over the horizon:

$$\begin{aligned} \sum_{t=0}^{T} r_t &= \sum_{t=0}^{T}\left[\frac{R_{\text{total}}(\mathbf{p}_t) - R_{\text{total}}(\mathbf{p}_{\text{SCA}})}{R_{\text{scale}}} - \lambda\|\boldsymbol{\delta}_t\|\right] \\ &= \frac{1}{R_{\text{scale}}}\sum_{t=0}^{T}[R_{\text{total}}(\mathbf{p}_t) - R_{\text{total}}(\mathbf{p}_{\text{SCA}})] - \lambda \sum_{t=0}^{T}\|\boldsymbol{\delta}_t\|. \end{aligned} \tag{9}$$

Since $\mathbf{p}_t = \mathbf{p}_{\text{SCA}} + \boldsymbol{\delta}_t$, consecutive differences telescope. Taking expectations yields the result. When corrections are small ($\lambda \to 0$), reward aligns directly with throughput improvement. □

Theorem 2 (Performance Floor Guarantee)**.** *For any policy π_θ with the floor mechanism in (7), the achieved throughput satisfies*

$$R_{\textit{total}}(\mathbf{p}_{\textit{final}}) \geq R_{\textit{total}}(\mathbf{p}_{\textit{SCA}}) \tag{10}$$

with probability one, for all parameter values θ.

Proof. The floor mechanism defines $\mathbf{p}_{\text{final}}$ as a conditional selection. Consider the two cases:

Case 1: If $R(\mathbf{p}_{\text{SCA}} + \boldsymbol{\delta}) \geq R(\mathbf{p}_{\text{SCA}})$, then $\mathbf{p}_{\text{final}} = \mathbf{p}_{\text{SCA}} + \boldsymbol{\delta}$, and by the condition itself, $R(\mathbf{p}_{\text{final}}) \geq R(\mathbf{p}_{\text{SCA}})$.

Case 2: If $R(\mathbf{p}_{\text{SCA}} + \boldsymbol{\delta}) < R(\mathbf{p}_{\text{SCA}})$, then $\mathbf{p}_{\text{final}} = \mathbf{p}_{\text{SCA}}$, yielding $R(\mathbf{p}_{\text{final}}) = R(\mathbf{p}_{\text{SCA}})$.

In both cases, $R(\mathbf{p}_{\text{final}}) \geq R(\mathbf{p}_{\text{SCA}})$. Since this holds deterministically for any realization of $\boldsymbol{\delta}$, the guarantee holds with probability one. □

Theorem 3 (Lyapunov Stability and Convergence)**.** *Assume $R_{\textit{total}}(\mathbf{p})$ is twice continuously differentiable with $\nabla^2 R_{\textit{total}}(\mathbf{p}) \preceq -\mu_R \mathbf{I}$ in a neighborhood of $\mathbf{p}^*$ (local strong concavity). Define the Lyapunov function*

$$\mathcal{L}(\mathbf{p}) = \|\mathbf{p} - \mathbf{p}^*\|^2. \tag{11}$$

Under gradient-based policy updates with step size $\alpha < 1/L$ where L is the Lipschitz constant of $\nabla R_{\textit{total}}$, the iterates satisfy

$$\mathcal{L}(\mathbf{p}_{t+1}) \leq (1 - \alpha\mu_R)^2 \mathcal{L}(\mathbf{p}_t), \tag{12}$$

implying exponential convergence: $\|\mathbf{p}_t - \mathbf{p}^*\| \leq (1 - \alpha\mu_R)^t \|\mathbf{p}_0 - \mathbf{p}^*\|$.

Proof. The policy gradient step moves in the direction of increasing throughput:

$$\mathbf{p}_{t+1} = \mathbf{p}_t + \alpha \nabla R_{\text{total}}(\mathbf{p}_t). \tag{13}$$

We analyze the Lyapunov function decrease:

$$\begin{aligned} \mathcal{L}(\mathbf{p}_{t+1}) &= \|\mathbf{p}_{t+1} - \mathbf{p}^*\|^2 \\ &= \|\mathbf{p}_t + \alpha\nabla R_{\text{total}}(\mathbf{p}_t) - \mathbf{p}^*\|^2 \\ &= \|\mathbf{p}_t - \mathbf{p}^*\|^2 + 2\alpha\langle \nabla R_{\text{total}}(\mathbf{p}_t), \mathbf{p}_t - \mathbf{p}^*\rangle \\ &\quad + \alpha^2 \|\nabla R_{\text{total}}(\mathbf{p}_t)\|^2. \end{aligned} \tag{14}$$

By strong concavity of R_{total} near $\mathbf{p}^*$:

$$\langle \nabla R_{\text{total}}(\mathbf{p}_t), \mathbf{p}_t - \mathbf{p}^*\rangle \leq -\mu_R \|\mathbf{p}_t - \mathbf{p}^*\|^2. \tag{15}$$

Using $\|\nabla R_{\text{total}}(\mathbf{p}_t)\| \leq L\|\mathbf{p}_t - \mathbf{p}^*\|$ from Lipschitz continuity:

$$\begin{aligned} \mathcal{L}(\mathbf{p}_{t+1}) &\leq (1 - 2\alpha\mu_R + \alpha^2 L^2)\|\mathbf{p}_t - \mathbf{p}^*\|^2 \\ &\leq (1 - \alpha\mu_R)^2 \mathcal{L}(\mathbf{p}_t) \end{aligned} \tag{16}$$

for $\alpha \leq \mu_R / L^2$. Iterating yields the exponential bound. □

Corollary 1 (Sample Efficiency)**.** *SGAC reaches ϵ-optimality in $\mathcal{O}(\log(1/\epsilon)/\mu_R)$ iterations, compared to $\mathcal{O}(1/\epsilon)$ for vanilla policy gradient without warm-start.*

TABLE I: Throughput Comparison (Mbps)

Method	Mean	Std	vs Random	vs SCA-20
Random	97.2	57.4	1.00×	0.27×
Static	97.7	37.9	1.01×	0.27×
Analytical	196.4	170.5	2.02×	0.54×
SCA-5	356.2	300.2	3.67×	0.97×
SCA-20	366.8	300.0	3.77×	1.00×
SCA-50	366.9	300.0	3.78×	1.00×
SGAC	**353.9**	**281.4**	**3.64**×	$\geq$**1.00**×

Proof. From Theorem 3, achieving $\|\mathbf{p}_t - \mathbf{p}^*\| \leq \epsilon$ requires

$$t \geq \frac{\log(\|\mathbf{p}_0 - \mathbf{p}^*\|/\epsilon)}{\log(1/(1-\alpha\mu_R))} \approx \frac{\log(1/\epsilon)}{\alpha\mu_R} \tag{17}$$

iterations. The SCA warm-start provides $\|\mathbf{p}_0 - \mathbf{p}^*\| = \mathcal{O}(1)$ rather than $\mathcal{O}(\text{diam}(\mathcal{P}))$ for random initialization, reducing the numerator. Standard RL without the throughput gradient in the state converges at rate $\mathcal{O}(1/\sqrt{t})$ in general, giving $\mathcal{O}(1/\epsilon^2)$ complexity. □

V. Experiments

A. Setup

We test on 50 randomly generated urban scenarios. Each has a base station at $(50, 50, 15)$ m and $K = 5$ users uniformly distributed in a 100×100 m area. The UAV altitude ranges from 10 to 40 m. Carrier frequency is 28 GHz with 100 MHz bandwidth.

B. Baselines

We compare against:

- **Random**: uniform sampling in the feasible region
- **Analytical**: weighted centroid of users and base station
- **Static**: fixed center position
- **SCA-5/20/50**: successive convex approximation with 5, 20, or 50 iterations
- **SGAC**: our method with floor guarantee

C. Results

Table I reports throughput averaged over test scenarios. SGAC achieves 353.9 Mbps, matching SCA-20 within measurement noise while guaranteeing $\geq$ SCA performance on every instance. Compared to random placement (97.2 Mbps), this is a 264% gain. Against the analytical heuristic (196.4 Mbps), the gain is 80%.

Convergence tests show SGAC reaches 95% of final performance in about 50 episodes, roughly four times faster than vanilla actor-critic. The floor guarantee held in all 50 test scenarios—SGAC never underperformed the SCA baseline.

Additional mobility tests with users changing position every 100 ms yielded average throughput of 385.1 Mbps, indicating robustness to dynamic conditions.

VI. Conclusion

We presented SGAC, a framework combining convex optimization with reinforcement learning for UAV relay positioning. The residual architecture and floor guarantee provide safety while allowing the agent to learn improvements beyond classical methods. Experiments confirm large throughput gains over heuristics and convergence rates several times faster than standard RL. Future directions include multi-UAV coordination and real-time trajectory adaptation.